Magnetic Grammar: A Bugfix

Marc van Oostendorp
Radboud University

1. Introduction

In D'Alessandro & Van Oostendorp (2020), we proposed *Magnetic Grammar*, a model of phonological competence in which a language's segment inventory is characterised entirely by features that *attract* or *reject* other features within a segment. The paper referred to a Python implementation demonstrating the learning algorithm. This squib reports on a bug in that implementation and presents a corrected version, which leads to a small but theoretically meaningful refinement of the learning algorithm.

The corrected implementation is available as an open-source repository at https://github.com/fonolog/MagneticGrammar.

2. The framework in brief

In Magnetic Grammar, segments are sets of privative features. Each feature F in a language can have two types of properties with respect to another feature G:

> **Attract(G):** F is satisfied only if G is *present* in the segment.
>
> **Reject(G):** F is satisfied only if G is *absent* from the segment.

A segment is well-formed if and only if all properties of all its features are satisfied. A word is well-formed if all its segments are. The set of features and their properties constitutes a complete characterisation of a language's phonology (and syntax, but the implementation was only about phonology) at the level of segment structure.

3. The bug

The original implementation contained a subtle error in the function that checks whether a feature is present in a segment. Due to this error, the attract and reject mechanisms were effectively inert: reject never actually blocked any segment, and attract never actually required anything. The published results – which were correct – seemed an artefact of this bug, since the effective behaviour of the system reduced to: *a segment is valid if and only if all its features are known to the grammar.*

This matters because the *learning algorithm* described in the paper depends crucially on attract and reject working correctly. The learner is supposed to posit attract and reject properties for new features and then prune contradicted properties against observed segments. With the bug, pruning always eliminated all attracts and kept all rejects — but since rejects never blocked anything, the result was equivalent to simply recording which features exist.

4. The fix

Fixing the evaluation functions is straightforward. The more interesting consequence is that the learning algorithm itself needs to be refined. The original algorithm assumed that when a new feature is encountered, it should attract *and* reject all other known features, with contradicted properties pruned during validation. With correct attract/reject evaluation, this leads to a problem: features that enter the grammar simultaneously (from the same first observed segment) become permanently bound by mutual attract constraints, because they have only ever been observed together.

The corrected algorithm rests on two principles:

> **(i) Co-new features get no mutual constraints.** When multiple features enter the grammar from the same segment, they receive no attract or reject properties relative to each other. There is no prior context to form hypotheses against.
>
> **(ii) Reject requires accumulated evidence.** A feature F only rejects G after F has been observed in two or more distinct segments and G was never present in any of them. A single observation is insufficient evidence for rejection.

New features do receive attract properties for previously known features that co-occur in the current segment. These attracts are pruned in the standard way when counter-evidence is observed.

5. Illustrations

Generalisation from /s/ and /p/. Suppose the learner observes /s/ (with features Anterior, Consonantal, Continuant, Coronal, Strident) and then /p/ (Anterior, Consonantal, Labial). After /s/, all five features enter the grammar with no properties (principle i). After /p/, the new feature Labial attracts the co-occurring known features Anterior and Consonantal. The grammar now predicts that /t/ (Anterior, Consonantal, Coronal) is valid — since Coronal has no constraints — and /f/ (Anterior, Consonantal, Continuant, Strident, Labial) is valid — since Labial's attract requirements are satisfied. The predicted inventory is {s, p, t, f}, matching the generalisation discussed in D'Alessandro & Van Oostendorp (2020).

Emergence of /v/. Continuing the learning path with /z/, /b/, and /f/, the system accumulates evidence. After observing /b/ (Anterior, Consonantal, Labial, Voice), the feature Voice has been seen in two segments (/z/ and /b/) and the grammar begins to develop reject constraints (e.g., Coronal rejects Labial after being observed twice without it). After /f/ is also learned, the system predicts /v/ (Anterior, Consonantal, Continuant, Strident, Labial, Voice): all features are known, Labial's attracts are satisfied, and no reject blocks the combination. The voiced labiodental fricative emerges without ever having been observed.

6. Conclusion

The corrected implementation reveals that the learning algorithm requires a more nuanced treatment of initial hypotheses than originally described. The two principles — no constraints between co-new features, and evidence-based reject — are not ad hoc patches but follow from a basic epistemological consideration: hypotheses about feature relationships require a comparative context. A feature observed for the first time provides no basis for comparison, and a single observation provides no basis for rejection.

The full implementation, including an interactive Jupyter notebook to learn and test Magnetic Grammar, uses the panphon library (Mortensen et al. 2016) for IPA-to-feature mapping and is freely available at https://github.com/fonolog/MagneticGrammar.

References

D'Alessandro, R. & M. van Oostendorp (2020). Magnetic Grammar. Linguistic Analysis, 42 3-4: 405-439. Available online

Mortensen, D. R., P. Littell, A. Bharadwaj, K. Goyal, C. Dyer & L. Levin (2016). PanPhon: A resource for mapping IPA segments to articulatory feature vectors. *Proceedings of COLING 2016*, pp. 3475–3484.